MIL-C-19262A(OS)

15 November 1976

SUPERSEDING
MIL-C-19262(NOrd)
3 January 1956
(see section 6)

MILITARY SPECIFICATION

CLOTH, PARACHUTE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification covers 12 ounce Nylon and Saponified acetate parachute cloth.

1.2 Classification. - The parachute cloth shall be of the following types, classes, and forms, as specified (see 6.2):

Type I. - Nylon
Type II. - Saponified acetate

Class C. - 12 ounce (see 6.3)

Form 1. - One single-body width per full width of cloth.
Form 2. - Multiple single-body widths per full width of cloth.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, Md. 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 8305

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SPECIFICATIONS

FEDERAL

PPP-P-1133	Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-851	Coding; Manufacturer's Color, Nylon Parachute Cloth

FEDERAL

FED-STD-4	Glossary of Fabric Imperfections
FED-STD-191	Textile Test Methods
FED-STD-595	Colors

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Yarn requirements.

3.1.1 Types. - The specified type of yarn used in the manufacture of the cloth shall be of the following types:

3.1.1.1 Type I. - Type I yarn shall be a bright, high tenacity, light and heat resistant polyamide prepared from hexamethylene diamine and adipic acid or its derivatives. Physical characteristics of the yarn shall be, as follows:

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a. - Melting point. The yarn's melting point shall be 482 ± 10°F (250 ± 6°C).
b. - Filament size. The yarn shall be a multifilament nylon of a denier and size to meet the cloth requirements of this specification.
c. - Bleaching. The yarn shall not be bleached in any manner.
d. - Age. Yarn more than two years old shall not be used in the manufacture of the cloth.

3.1.1.2 Type II. - The acetate yarn shall be a saponified oriented acetate yarn. Physical characteristics of the yarn shall be as follows:

a. - Filament size. The yarn shall be of continuous multifilament of a size to meet the cloth requirements of this specification.
b. - Age. Yarn more than one year old shall not be used in the manufacture of the cloth.

3.2 Cloth requirements.

3.2.1 Classes (Weights).

3.2.1.1 Class C. - The weight, in ounces per square yard, of Class C cloth shall be as given in Table I.

3.2.2 Forms. - The cloth shall be woven in the following forms, as specified in the contract or purchase order:

3.2.2.1 Form 1. - The cloth shall have one single-body width with selvages as shown in Figure 2.

3.2.2.2 Form 2. - The cloth shall have two or more specified single-body widths per the width of cloth as shown in Figure 1. The construction of Form 2 cloth shall be as follows:

a. - Dupe selvage edges shall be constrained by a minimum of 1 pair of leno locked ends.
b. - Separations linking the single-body widths shall be continuations of the fill yarn.

3.2.3 Weave. - The patterns shall be as shown in Figure 3.

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Table I. - Physical properties of the cloth.

Property	Type I Cloth	Type II Cloth
Weight; in oz/sq. yd.		
(a) Natural color	12 ± 0.5	12 ± 0.5
(b) Colored	13 ± 0.5	13 ± 0.5
Width, single-body; in inches (Body & 2 selvages/ dupe-selvages)	11 ± 1/4	11 ± 1/4
Width, each selvage or dupe-selvage; in inches	1-1/8±1/8	1-1/8±1/8
Width, each separation; in inches	1/2 ± 1/8	1/2 ± 1/8
Breaking strength, body; in lbs. per inch of width, min.		
(a) Warp direction	625	500
(b) Fill direction	425	330
Breaking strength, selvage or dupe-selvage; in lbs. per inch of width, min.		
(a) Warp direction	950	775
Elongation, body; in % min.		
(a) Warp, direction	30	10
(b) Fill direction	30	10
Tear resistance, body; in lbs., min		
(a) Across warp	75	70
(b) Across fill	60	55
Air permeability, body; in cu.ft./minute/sq.ft.	70 ± 20	70 ± 20

3.2.4 Colors.

3.2.4.1 Natural. - Unless otherwise specified, the color of the cloth shall be natural.

3.2.4.2 Specified colors. - When specified, the color of the cloth shall conform with the standard color number selected from FED-STD-595.

3.2.4.3 Camouflage. - When specified, the camouflage pattern and colors shall conform with those of the sample or instructions supplied by the procuring agency.

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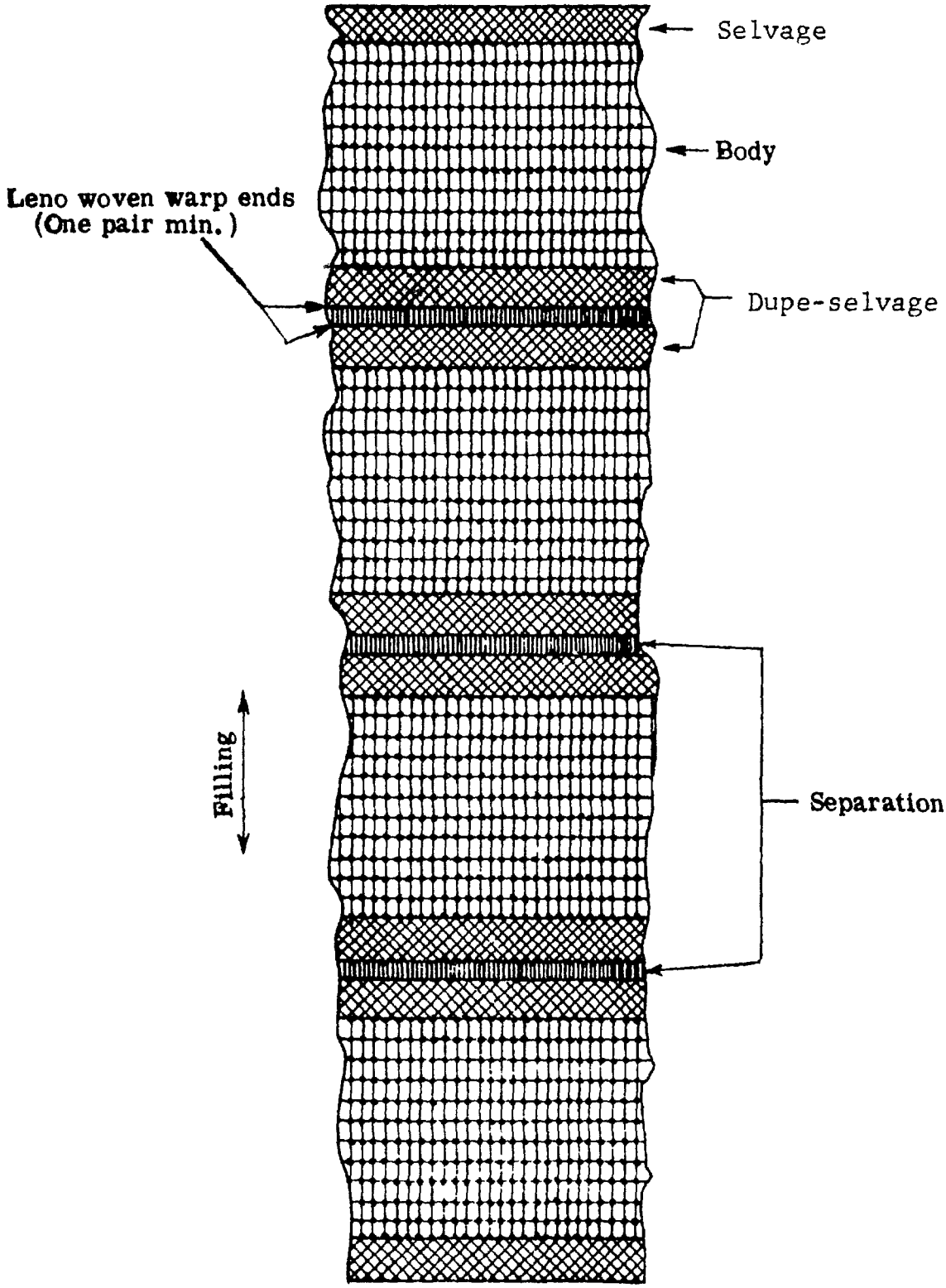


Figure 1.- Form 2 Cloth

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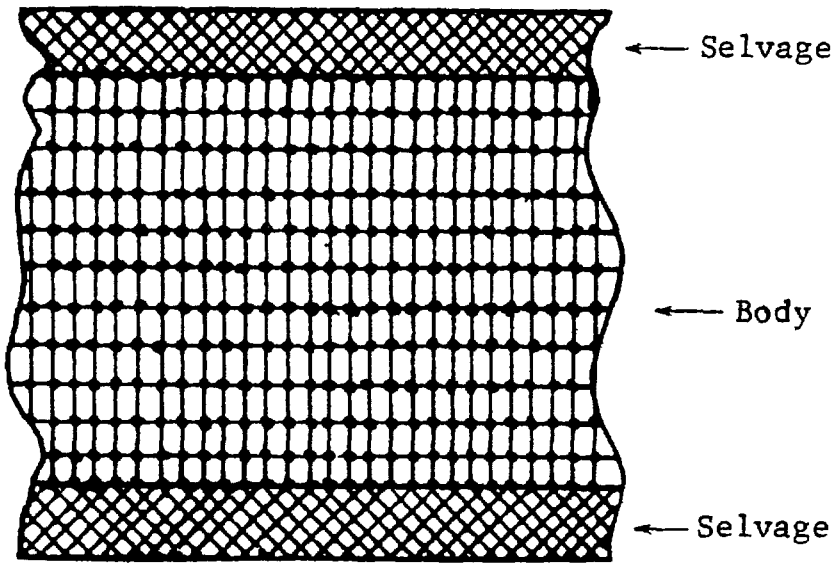


Figure 2.- Form 1 Cloth.

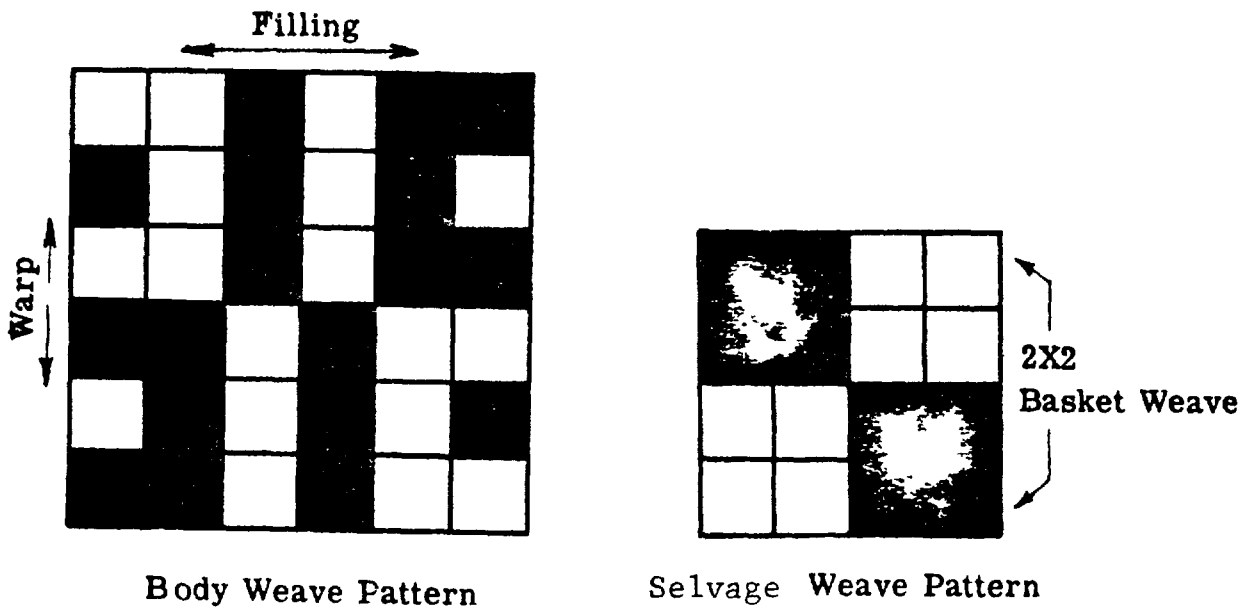


Figure 3.- Weave Patterns.

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3.2.4.4 Color matching. - The specified color or colors shall match the approved standard under natural (north sky) daylight or artificial daylight having a color temperature of 7,500° kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2,800° kelvin.

3.2.4.5 Colorfastness. - Colors shall show good colorfastness to salt water.

3.2.5 Finish. - The finished cloth shall be smooth and even and shall not contain sizing, lubricating, or weighting materials.

3.2.5.1 Acidity. - The pH of the water extract shall be in the range from 4.5 to 8.5.

3.2.6 Physical properties. - The cloth shall conform to the applicable requirements of Table I. Testing shall be in accordance with Table V.

3.2.6.1 Stability. - When subjected to the stability test, the cloth shall conform to the following criteria:

a. - Air permeability shall be within the limits specified in Table I.
b. - The thickness of the cloth shall not increase more than 10%.
c. - Shrinkage of the cloth shall not exceed 2% in the warp and 1% in the filling.

3.3 Dimensions.

3.3.1 Width dimensions. - Allowable widths and tolerances for single-body selvage, dupe-selvage, and separations shall be as shown in Table I.

3.3.2 Lengths. - Unless otherwise specified, the cloth shall be continuous pieces not less than 100 yards. Shorter pieces may be included in the total order in accordance with the following:

75% in pieces of 100 yards or more.
15% in pieces of 25 to 100 yards.
10% in pieces of 15 to 25 yards.

3.3.2.1 Rolls. - Unless otherwise specified, from 100 to 200 yards of cloth shall be put up on each roll. The pieces shall be put up on the rolls in accordance with PPP-P-1133.

3.3.2.2 Short Lengths. - Lengths shorter than 50 yards shall be rolled together to the specified roll size and the roll labeled to indicate that it is composed of short lengths.

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3.4 Defects. - The following defects, as defined in FED-STD-4 and common usage in commercial practice, will be allowed up to a maximum of 5 defects per 100 yards:

a. - Broken picks
b. - Broken or missing ends
c. - Floats
d. - Holes
e. - Mispicks
f. - Bowed fillings (see 3.4.1)
g. - Smashes
h. - Oil spots
i. - Thin areas due to starting or stopping of the loom or to pickouts.
j. - Torn, wavy, or tight selvage.

3.4.1 Bowed filling. - A bowed filling is a variation of the filling yarns from the normal, in the form of a single or compound arc. A bowed filling is classed as a defect when the perpendicular distance from the highest point on the arc to the straight line chord of the arc, in the plane of the cloth, exceeds 1 inch. The presence of a false bow or filling distortion that can be eliminated within the above requirements by straightening the cloth by hand when unrolled on a flat surface will not be considered a defect.

3.4.2 Marking of defects. - Cloth submitted for Government inspection shall indicate the presence of each defect by a red string sewn in the selvage opposite the defect.

3.4.3 Allowance for defects. - The total yardage supplied under a contract or order, shall include at least 1/2 additional yard for each marked defect. For a continuous defect extending over a length greater than 1/2 yard (such as: a broken warp end, float, or torn, wavy, or tight selvage), the additional yardage allowed shall equal the length of the defect.

3.5 Identification.

3.5.1 Fiber identification. - Each roll of cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the rules and regulations under the Textile Fiber Products Identification Act.

3.5.2 Manufacturer's (weaver's) identification. - Colored threads shall be woven into the selvage edges of the natural colored cloth to identify the weaver of the cloth to the procuring activity. The colors used shall be those indicated in MIL-STD-851. The dye used in the marker threads shall have no deleterious effect on the physical properties of the thread and the dyed thread shall show no bleeding or color transfer when subjected to the tests for laundering and dry cleaning as specified in Table IV.

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3.5.3 Identification of product. Each roll of finished cloth shall be marked for identification in accordance with PPP-P-1133. In addition, each piece of cloth in each roll shall be clearly and legibly marked with the finisher's roll number or code, and each roll shall have attached a durable tag on which the finisher's roll number or code is listed. The date of manufacture of the cloth shall be included on the tag attached to each roll.

3.6 Workmanship. The finished cloth shall be clean and evenly woven and shall conform to the quality and grade of product established by this specification. The finished cloth shall be free from defects, except to the extent specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot inspections. Each lot of cloth submitted for acceptance inspection (4.3) shall be subjected to the following inspections:

a. Yarn testing (4.3.1)
b. Yard-by-yard examination (4.3.2)
c. Roll-by-roll examination (4.3.3)
d. Lot testing for physical properties (4.3.4)

4.2.1 Inspection lot. - An inspection lot shall consist of those finisher's rolls of cloth of one type, class, and form which were manufactured from the same lot of yarn, under essentially the same conditions and presented for inspection at the same time.

4.2.2 Unit of product.- For procurement purposes (6.2) the unit of product is a linear yard of a single-body width. For inspection purposes, a unit of product is either a linear yard or a roll of product as specified in 4.3.1 through 4.3.4.

4.3 Inspection procedures.

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4.3.1 Yarn testing. - Each roll of yarn shall be tested for compliance to the characteristics of paragraphs 3.1.1.1 or 3.1.1.2 as applicable (see Table II). Melting point (3.1.1.1) shall be tested twice per roll according to FED-STD-191, Test Method 1534 with results reported in °F or °C. A contractor's certificate of compliance to all other characteristics with the filament size reported in "Denier" and the age reported in "months" is required. Acceptance of the lot shall be in accordance with MIL-STD-105, Level S-3, and an AQL of 1.0% for all characteristics considered as a class.

Table II. - Yarn tests

Cloth Type	Characteristic	R'qmt Para	Test Method (2)	Tests per Unit of Product	Results Reported As
Type I	Chemical identity	3.1.1.1	(1)	1	Pass/Fail
	Brightness	3.1.1.1	(1)	1	Pass/Fail
	Tenacity	3.1.1.1	(1)	1	Pass/Fail
	Light Resistance	3.1.1.1	(1)	1	Pass/Fail
	Heat Resistance	3.1.1.1	(1)	1	Pass/Fail
	Melting Point	3.1.1.1	1534	2	°F or °C
	Filament size	3.1.1.1	(1)	1	Denier
	Unbleached	3.1.1.1	(1)	-	Pass/Fail
	Age	3.1.1.1	(1)	-	Months
Type II	Chemical identity	3.1.1.2	(1)	1	Pass/Fail
	Filament size	3.1.1.2	(1)	1	Denier
	Age	3.1.1.2	(1)	1	Months

Notes: (1) Contractor's certificate of compliance to this characteristic is required.
(2) Test Method numbers are those of FED-STD-191.

4.3.2 Yard-by-yard examination. - The cloth shall be examined for the fabric defects listed in section 3.4. From the lot of cloth, rolls shall be randomly selected for this examination in the quantity specified in Table III. From each of these sample rolls, an approximately equal number of yards shall be visually examined. The unit of product for this examination shall be one yard. Lot acceptance shall be allowed with a maximum of 5 defects per 100 yards of each sample roll. Additional yardage allowances for defects in acceptable lots, shall be supplied in accordance with 3.4.3.

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Table III. - Sampling of rolls for yard-by-yard examinations

Lot Size (Yards)	No. of Sample Rolls (1)
3200 or less	8
3201 to 10000, incl	13
10001 and over	20

Note: (1) Only one roll shall be taken from a shipping container. If the number of shipping containers is less than the the required sample size, a roll shall be taken from each container.

4.3.3 Roll-by-roll examination. - Each of the rolls sampled in accordance with 4.3.2 shall be examined overall for the defects listed in Table IV. The unit of product shall be one roll. Except as specified in 4.3.3.1 and 4.3.3.2, any roll having one or more Major defects or a combination of one Major defect plus two or more Minor defects shall be classed as defective and be cause for rejection of the lot.

Table IV. - Roll-by-roll examination

Requirement	Requirement Para or Reference	Classification of Defect	
		Major	Minor
Form	3.2.2	X	
Width, single-body	3.3.1	X	
Width, selvage	3.3.1		X
Width, dupe-selvage	3.3.1		X
Width, separations	3.3.1		X
Color, natural	3.2.4.1		X
Color, specified	3.2.4.4		X
Color, camouflage	3.2.4.3		X
Finish, smooth and even	3.2.5	X	
Finish, no sizing	3.2.5	X	
Finish, no lubricant	3.2.5	X	
Finish, no weighting	3.2.5	X	
Length, individual pieces	3.3.2	X	
Length, individual roll	3.3.2.1	X See 4.3.3.1	
Length, total rolls	Contract	X See 4.3.3.2	
Identification of short lengths	3.3.2.2	X	
Fiber identification	3.5.1	X	
Manufacturer's identification	3.5.2		X
Product identification	3.5.3		X
Colorfastness to laundering and dry cleaning (1)	3.5.2		X
Cleanness	3.6	X	

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Table IV. - Roll-by-roll examination (contd)

Note: (1) Acceptance with respect to this characteristic will be based on the contractor's certificate of compliance.

4.3.3.1 Individual roll yardage. - Any roll length less than the minimum specified or more than 2 yards below the length marked on the roll's ticket shall be classed as defective. Two or more defective rolls shall be cause for rejection of the lot.

4.3.3.2 Total roll yardage. - If the total of the lengths for all rolls measured in 4.3.3.1 does not conform to the totals obtained from tickets on the rolls, the lot shall be classed as defective and shall be cause for rejection of the lot.

4.3.4 Lot testing for physical properties. - The cloth shall be tested for its physical properties in accordance with Table V. The unit of product shall be one linear yard of cloth and the lot size shall be determined as the number of units of product in the lot. Each test sample shall consist of a continuous 3-yard piece of cloth, full width of material, which shall be randomly selected from a separate roll making up the inspection lot. The test sample size shall be in accordance with MIL-STD-105 Level S-2 and its acceptability shall be an AQL of 1.5% defective.

4.4 Test procedures.

4.4.1 Stability. - A sample 20 inches square shall be conditioned for 4 hours in an atmosphere having a relative humidity of 65± 2 percent and a temperature of 70° ± 2°F. The sample shall be measured for thickness in accordance with Method 5030 of FED-STD-191, marked with an 18-inch square and then formed into a loop by stapling two opposite sides together. A glass rod 1/2 inch in diameter and 21 inches long shall be placed inside the loop at the top, and a glass rod 1/4 inch in diameter and 21 inches long, weighing approximately 45 grams, shall be placed inside the loop at the bottom. The sample shall be immersed in a tub of boiling water by attaching twine or wire to the ends of the top glass rod and allowing the sample to hang freely in the boiling water for 15 minutes. The sample shall then be removed and allowed to drain. After the samples are removed the sample shall be placed in a flat position on a horizontal screen and thoroughly dried. It shall then be conditioned for 4 hours in an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of 70° ± 2°F. The marked square shall be measured in three places in the warp direction and three places in the filling direction and the amount of shrinkage determined. Air permeability tests, as described in Method 5450 of FED-STD-191, shall be performed and the thickness shall again be measured. Changes in the characteristics of the cloth shall not be in excess of the amounts specified in 3.2.6.1.

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Table V. - Tests for physical properties.

Test Characteristic	Requirement Reference	Test Method (1)
Weave	3.2.3	Visual
Weight	3.2.1.1	5041
Colorfastness to salt water	3.2.4.5	5632
Acidity, pH	3.2.5.1	2811
Breaking strength, body	3.2.6	5104
(a) Warp direction		
(b) Fill direction		
Breaking strength, selvage or dupe selvage	3.2.6	5104
(a) Warp direction		
Elongation, body	3.2.6	5104
(a) Warp direction		
(b) Fill direction		
Tear resistance	3.2.6	5134
(a) Across warp		
(b) Across fill		
Air permeability, body	3.2.6	5450
Stability	3.2.6.1	4.4.1
(a) Air permeability		5450
(b) Thickness increase		Visual
(c) Shrinkage		Visual

Note: (1) Test method numbers are those of FED-STD-191.

4.5 Examination for preparation for delivery. - Examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with.

5. PACKAGING

5.1 Packaging. - Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Levels A and C. - The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. - Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. - The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

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6. NOTES

6.1 Intended use. - The cloth is intended for use in the manufacture of ring-slot type parachutes used for Navy ordnance applications.

6.2 Ordering data. - Procurement documents should specify the following:

a. - Title, number, and date of this specification
b. - Type, class, and form of cloth required (see 1.2)
c. - Color (see 3.2.4)
d. - Yardage required (single body width)
e. - Level of packaging and packing required

6.3 Correlation data. - Class A and B cloths of MIL-C-19262(NOrd) are obsolete and have been dropped from MIL-C-19262A(OS). The Class C designation is carried in this specification to provide clarity for reprocurements and drawings.

6.4 Narrow looms. - Existing narrow fabric looms may be used to weave the cloth by folding over a width on the loom and, subsequently, unfolding the cloth of the loom to the required width.

6.5 Changes from previous issue. - Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing Activity:
Navy - OS

Project No.
8305-N590

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-C-19262A(OS) "Cloth, Parachute"

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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